Runbook: Stream PC Audio to Sonos Beam on a Mobile Hotspot (No Phone App)

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Problem

Sonos Beam cannot be discovered or streamed to when the only internet source is a mobile hotspot. Hotspots enforce client isolation, blocking device-to-device traffic. HDMI ARC/CEC on the TV (Insignia NS-40D510NA21) also failed handshake ("No CEC device found"). Sonos phone app not in use by policy.

Solution Summary

Bypass the hotspot entirely. Hardwire the Beam to the PC with an ethernet cable, share the PC's Wi-Fi connection to that ethernet port (Windows ICS), and stream system audio with swyh-rs in WAV format.

Hardware

- Windows 11 PC with ethernet port
- Sonos Beam
- Cat5/Cat5e/Cat6 cable, PC ethernet port to Beam ethernet port
- Internet: phone hotspot over Wi-Fi (Galaxy S24 Ultra)

Network Architecture

- Wi-Fi adapter: hotspot internet (10.238.x.x)
- Ethernet adapter: shared via ICS, PC becomes 192.168.137.1
- Beam: pulls DHCP from the PC, lands on 192.168.137.x
- swyh-rs streaming server: http://192.168.137.1:5901/stream/swyh.wav

Setup Steps

1. Connect ethernet cable: PC port to Beam port.
2. Win+R, run `ncpa.cpl`.
3. Right-click the Wi-Fi adapter > Properties > Sharing tab.
4. Check "Allow other network users to connect through this computer's Internet connection." Select Ethernet in the dropdown. OK. Accept the 192.168.137.1 notice.
5. Windows Firewall: Start > type "firewall" > Allow an app through Windows Firewall > Change settings > swyh-rs row > check BOTH Private and Public. OK. (The ICS

ethernet link shows as "Unidentified network" = Public profile. Private-only = silent block.)

6. swyh-rs > Network tab > Active network: 192.168.137.1. Apply configuration changes. Restart swyh-rs.
7. swyh-rs > Audio tab > streaming format: WAV 16-bit. Enable Inject Silence. Apply. Restart. (Sonos closes LPCM streams immediately. WAV is the documented Sonos-compatible format; Inject Silence prevents Sonos killing the stream during audio pauses. Ref: swyh-rs GitHub README and issue #71.)
8. Network tab > Run SSDP discovery now. Beam appears as "Sonos Beam Media Renderer - RINCON_xxxx".
9. Windows sound output: Speakers/Headphones (Realtek). swyh-rs captures this device.
10. Check the Beam's box in swyh-rs. Play audio.
11. Echo fix: mute the PC volume. swyh-rs captures pre-mute on this hardware; Beam keeps playing.

Troubleshooting

Symptom	Cause	Fix
Beam not in renderer list	Discovery stale	Network tab > Run SSDP discovery now
Beam missing after reboot	ICS sometimes resets	Recheck Wi-Fi Sharing tab, then rediscover
Stream starts then instantly stops, loops	LPCM format or firewall Public unchecked	WAV format + both firewall boxes
Beam plays then dies during quiet passages	Silence timeout	Enable Inject Silence
Echo/doubled audio	PC speakers live	Mute PC volume
Wrong network in swyh-rs (10.238.x.x)	Hotspot adapter selected	Select 192.168.137.1, apply, restart

Known Dead Ends (do not retry)

- HDMI ARC through this TV: CEC handshake fails, "No CEC device found" persists through INlink settings and audio reset.

- swyh-rs discovery on the hotspot network directly: blocked by client isolation, unfixable in software.
- Approx. 1-2 second latency is inherent to network streaming. Fine for music, not for video sync or gaming.

Screenshots to Attach

1. swyh-rs Status tab showing: Format wav 16 bit, Inject silence checked, Active network 192.168.137.1, and the Beam in the renderer list.
2. ncpa.cpl showing both adapters (Wi-Fi + Ethernet).
3. Wi-Fi Properties > Sharing tab with the box checked.
4. Firewall Allowed apps list, swyh-rs row, both boxes checked.